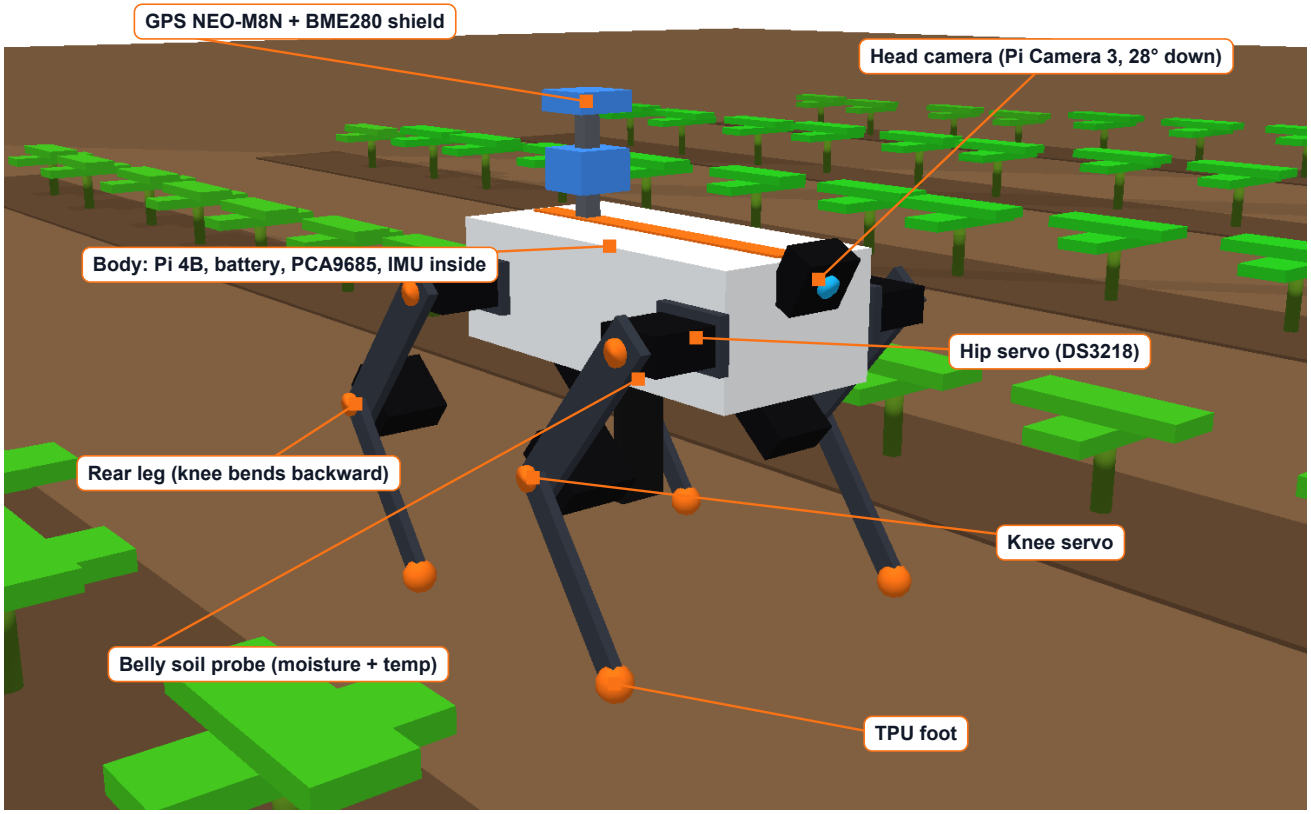
AgriRover-Q — Robot Figure (v0.2)

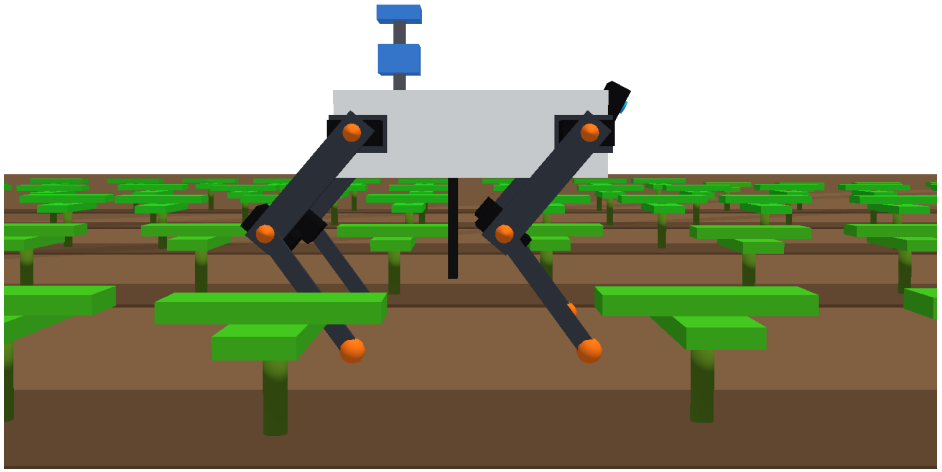
Spot-style quadruped for crop-field soil surveys. Renders from the simulation model (design/3d/agrirover_q.urdf); dimensions from agrirover/spec.py.



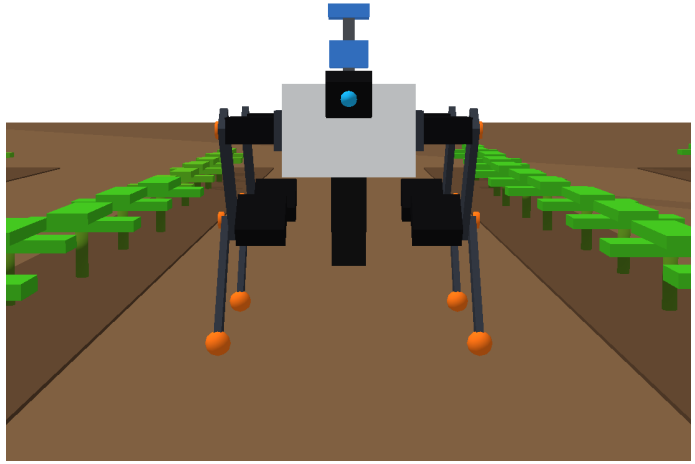
Size (walking)	303 × 217 × 282 mm
Body	220 × 100 × 70 mm
Legs	thigh 100 / shin 110 mm, 3 servos each
Mass	1.95 kg
Belly clearance	140 mm walking, 35 mm crouched
Probe depth	50 mm
Camera	28° down (was 40° in the approved 2D sheet)
Speed / runtime	~0.15 m/s, ~62 min

Changes since the approved 2D design (all found in simulation): camera pitch 40° → 28° because 40° saw only bare soil between the rows; walking-trot gait (duty 0.7, 0.5 s cycle) instead of a pure trot, which slipped and could fall over; photos taken standing so the crop rows are in frame. Page 2: poses. Page 3: dimensioned layout sheet.

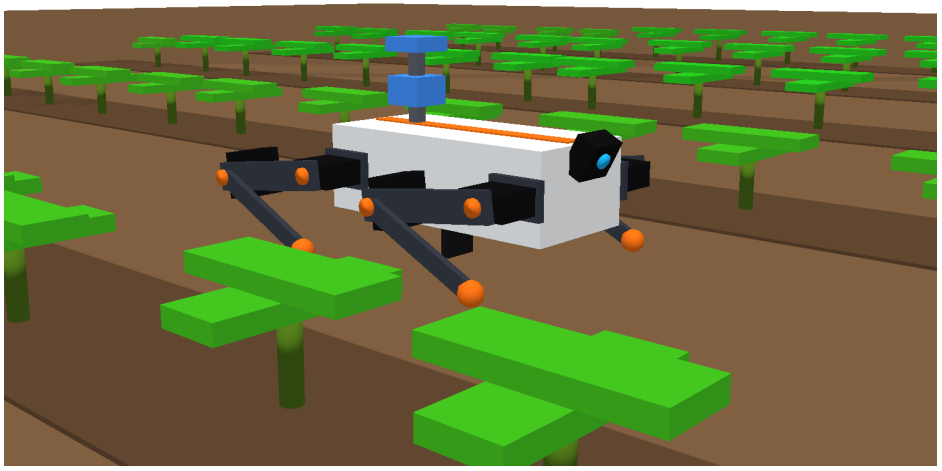
Poses



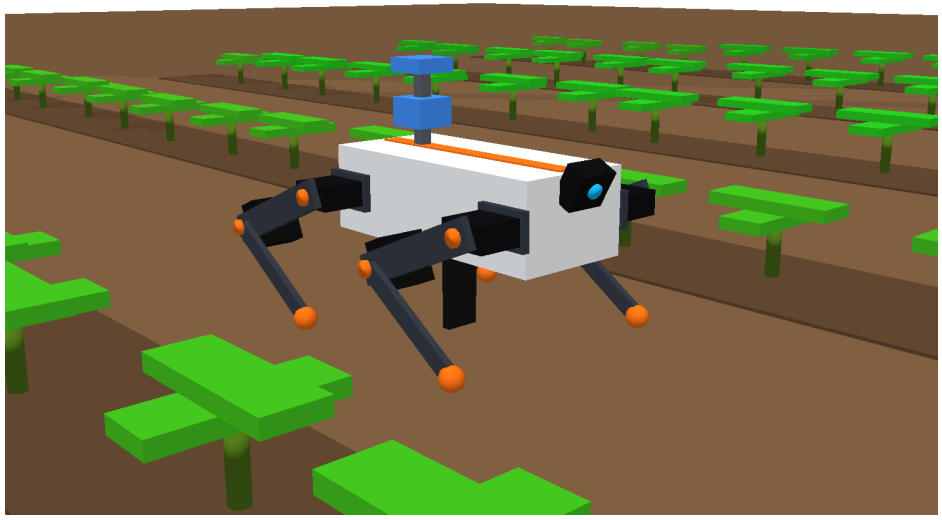
Walking: side view (all knees bend backward)



Walking: front view



Crouched to sample: belly at 35 mm, probe 50 mm into the soil



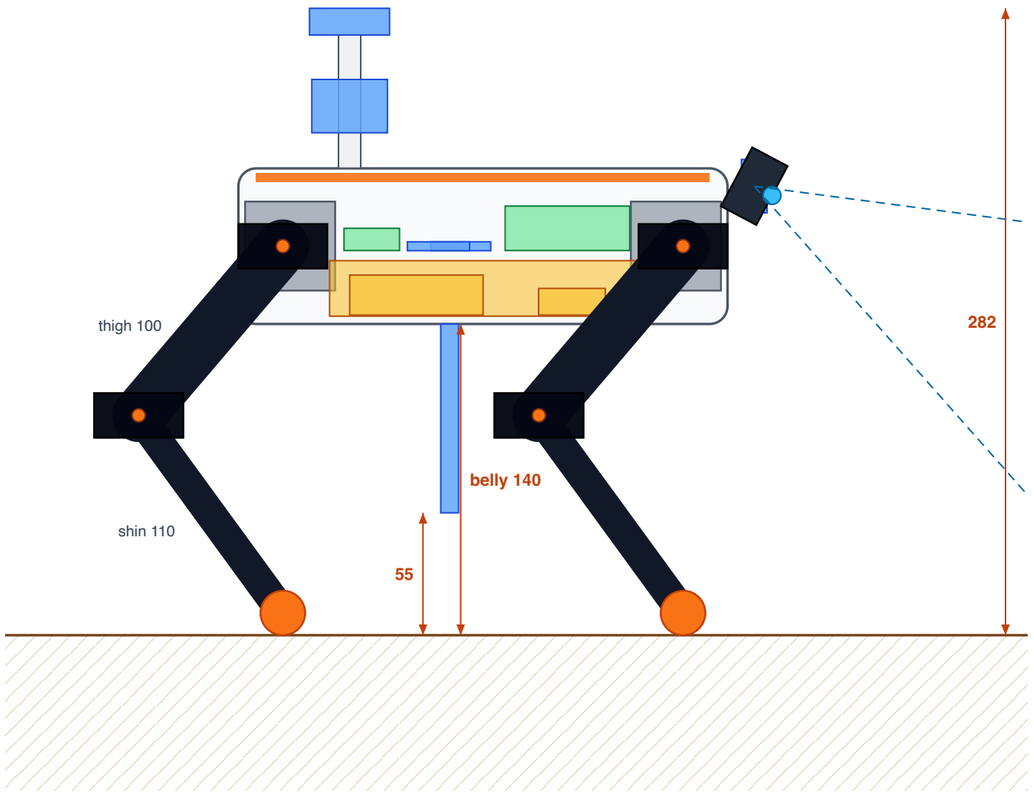
Sitting: parked / charging

AgriRover-Q v0.2 — quadruped layout (sim-tested)

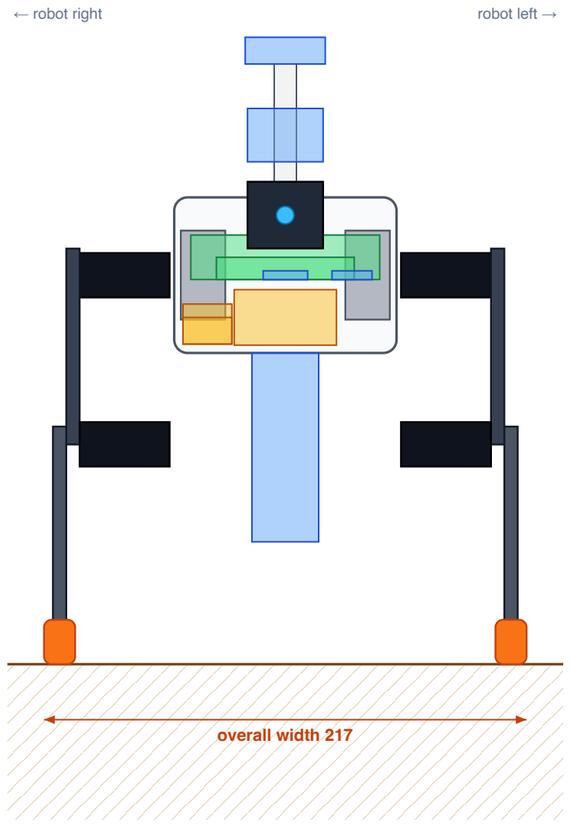
Spot-style dog robot · 12 servos · all dimensions mm · generated from agrirover/spec.py

SIDE — WALKING

near legs dark, far legs grey · camera view dashed

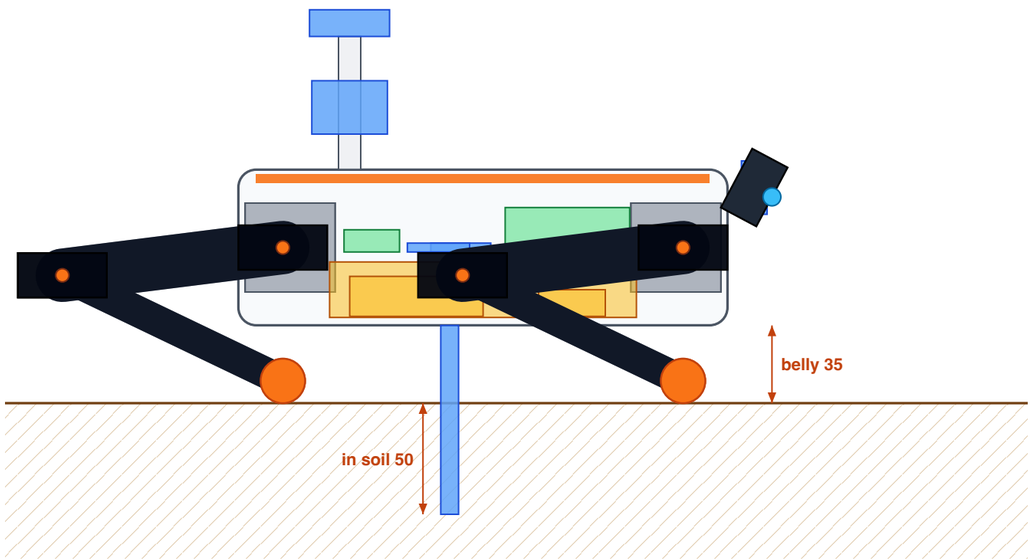


FRONT — WALKING



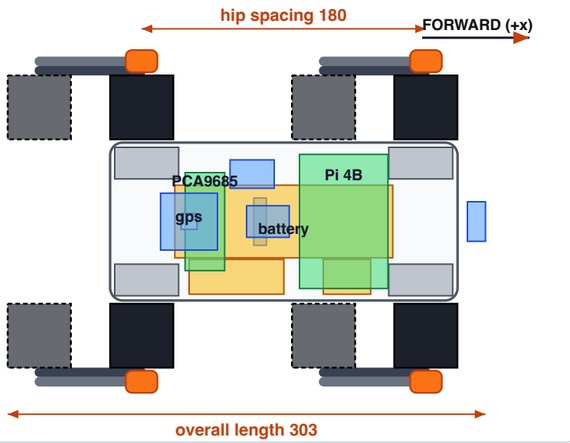
SIDE — CROUCHED TO PROBE

legs fold, body weight pushes the belly probe in



TOP — WALKING

knee servos (under the thigh) dashed



Key specs

Overall L × W × H (walking)	303 × 217 × 282
Body	220 × 100 × 70
Leg: thigh / shin	100 / 110
Belly clearance walking	**140 mm**
Probe tip clearance walking	55 mm
Probe depth when crouched	50 mm
Total mass (est.)	**1.95 kg**
Walking speed (trot)	~0.15 m/s
Runtime (est.)	**~62 min**
Joint torque need, 2 feet down	6.3 kg·cm
Servo stall torque	20 kg·cm (×3.2)
Camera height / pitch	202 mm / 28°
Camera sees ground	301–1656 mm ahead

Mass budget

servos ×12	720 g
leg structure ×4	412 g
battery	260 g
body shell (printed PETG)	230 g
wiring + fasteners	100 g
raspberry pi	61 g
servo buck	40 g
gps	25 g
soil probe	25 g
small parts	72 g
TOTAL	**1945 g**

Power budget

12 servos: walking 35 W × 60%	21.0 W
12 servos: holding pose 8 W × 40%	3.2 W
Pi 4B + camera + WiFi (via buck)	3.9 W
sensors (GPS, IMU, BME280, ADC, probe)	0.4 W
TOTAL average	**28.5 W**

Parts

12 × DS3218 servo	3 per leg
PCA9685	servo driver
Raspberry Pi 4B	brain
2S 5000 mAh LiPo	battery
6V 20A + 5V 5A bucks	power
BNO055	IMU / balance
NEO-M8N	GPS
BME280	air
Cap. probe + DS18B20	soil
Pi Camera Module 3	photos + rows

Design notes

- Legs instead of wheels: steps over clods, furrows and mud ridges; 140 mm belly clearance while walking.
- Spot-style: every knee bends backward. Direct-drive servos at shoulder, hip and knee (simple, no linkages).
- Soil probe is fixed under the belly. To sample, the robot crouches and its own weight pushes the probe 50 mm in.
- Camera head at the nose, 28° down; GPS + BME280 mast at the rear; battery low in the centre.
- Battery is now 2S 5000 mAh: 12 walking servos draw ~10× what the wheels did (2200 mAh would give ~25 min).